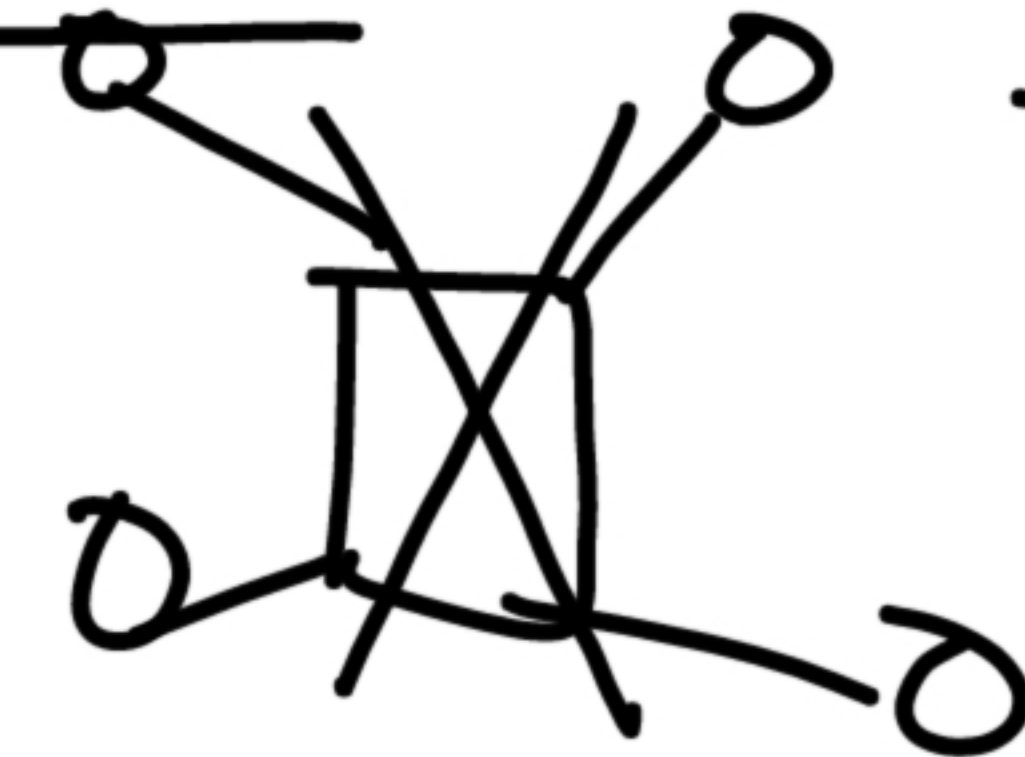
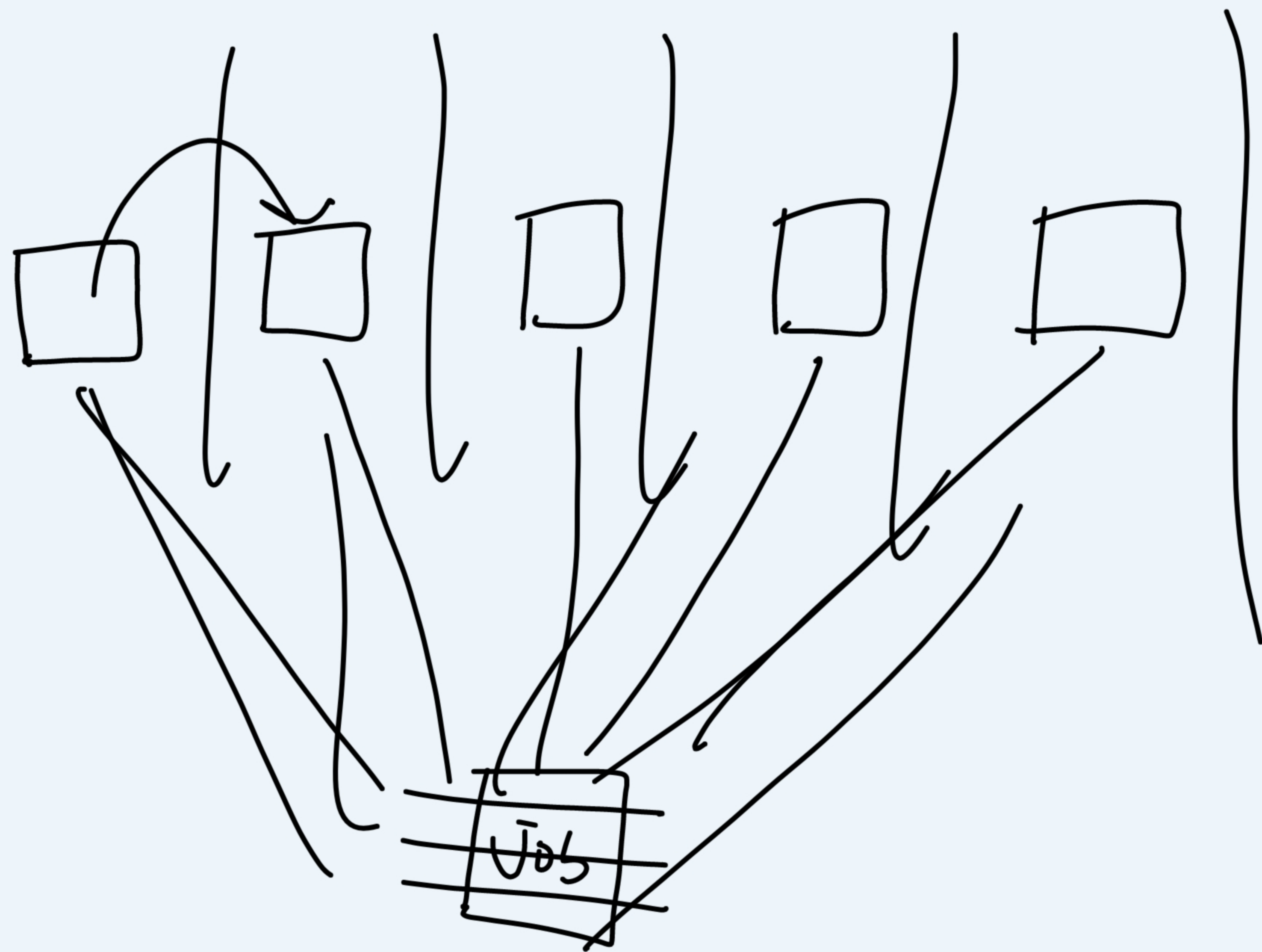


Distributed/Parallel/Computing vs Cloud Computing

Grid

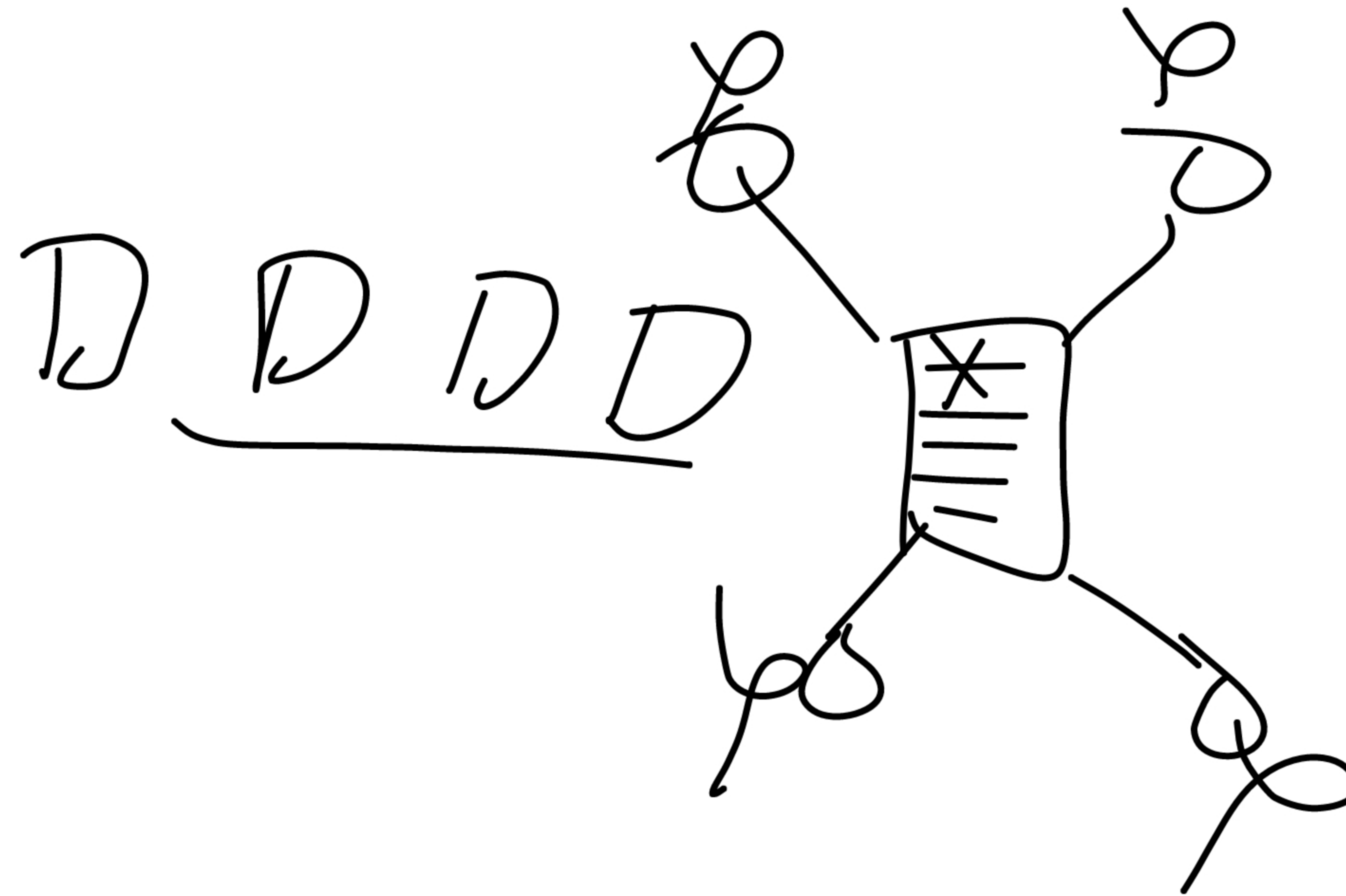
- Distributed Parallel Computing involves distributing tasks across multiple computers or servers to execute simultaneously (in parallel).
- The main goal is to break down large problems into smaller tasks that can be processed independently on separate systems.
- Cloud computing provides on-demand access to shared resources (like servers, storage, applications) over the internet, managed by a cloud service provider.
- Users can scale resources up or down based on their needs without worrying about underlying infrastructure.





Similarities

- Scalability
- Resource sharing
- Task distribution
- Fault tolerance



Aspect	Distributed Parallel Computing	Cloud Computing
Purpose	Solves large problems by splitting tasks for simultaneous processing	Provides access to computing resources, storage, and applications as services
<u>Infrastructure Ownership</u>	Usually owned and managed by the organization or research entity	Often managed by third-party cloud providers (like AWS, Azure, Google Cloud)
<u>Resource Management</u>	Requires <u>manual</u> configuration and task distribution	Automatic scaling and resource management via cloud provider
Deployment	Requires specialized hardware and setup for specific tasks	Available as a service, accessible from anywhere with internet
Use Cases ✓	High-performance computing, scientific simulations, data analytics ✓	Web <u>hosting</u> , storage solutions, SaaS applications, <u>data</u> backups

Grid Computers
Dis + Internet = Cloud-